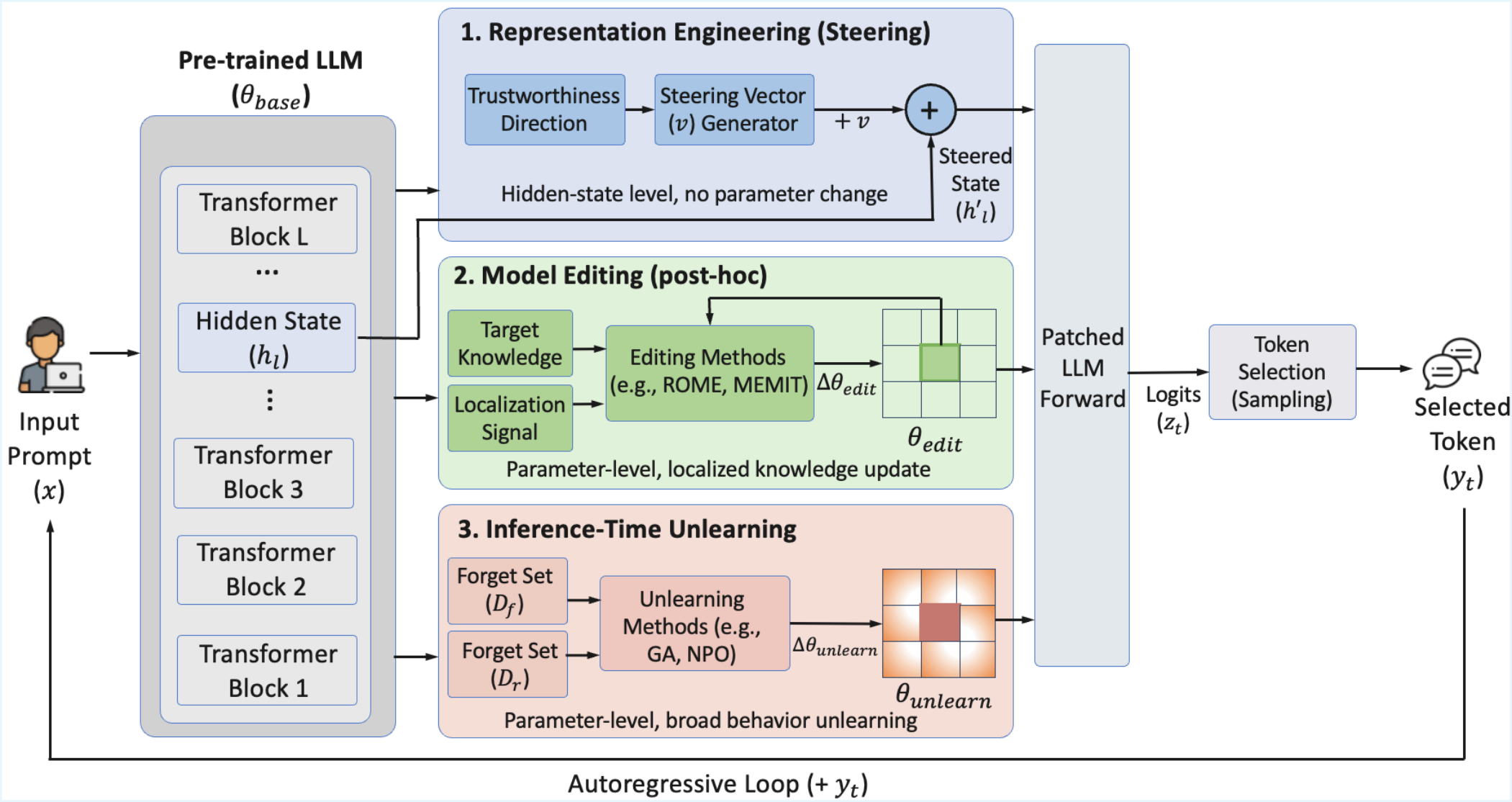

Pre-trained LLM
(θ_{base})
Transformer Block L
...
Hidden State (h_l)
⋮
Transformer Block 3
Transformer Block 2
Transformer Block 1
Input Prompt (x)
1. Representation Engineering (Steering)
Trustworthiness Direction
Steering Vector (v) Generator
$+v$
Steered State (h'_l)
Hidden-state level, no parameter change
2. Model Editing (post-hoc)
Target Knowledge
Localization Signal
Editing Methods (e.g., ROME, MEMIT)
$\Delta\theta_{edit}$
θ_{edit}
Parameter-level, localized knowledge update
3. Inference-Time Unlearning
Forget Set (D_f)
Forget Set (D_r)
Unlearning Methods (e.g., GA, NPO)
$\Delta\theta_{unlearn}$
$\theta_{unlearn}$
Parameter-level, broad behavior unlearning
Patched LLM Forward
Logits (z_t)
Token Selection (Sampling)
Selected Token (y_t)
Autoregressive Loop (+ y_t)